

MULTI-BRANCH OFFICE NETWORK PROJECT REPORT

Cisco Packet Tracer Simulation

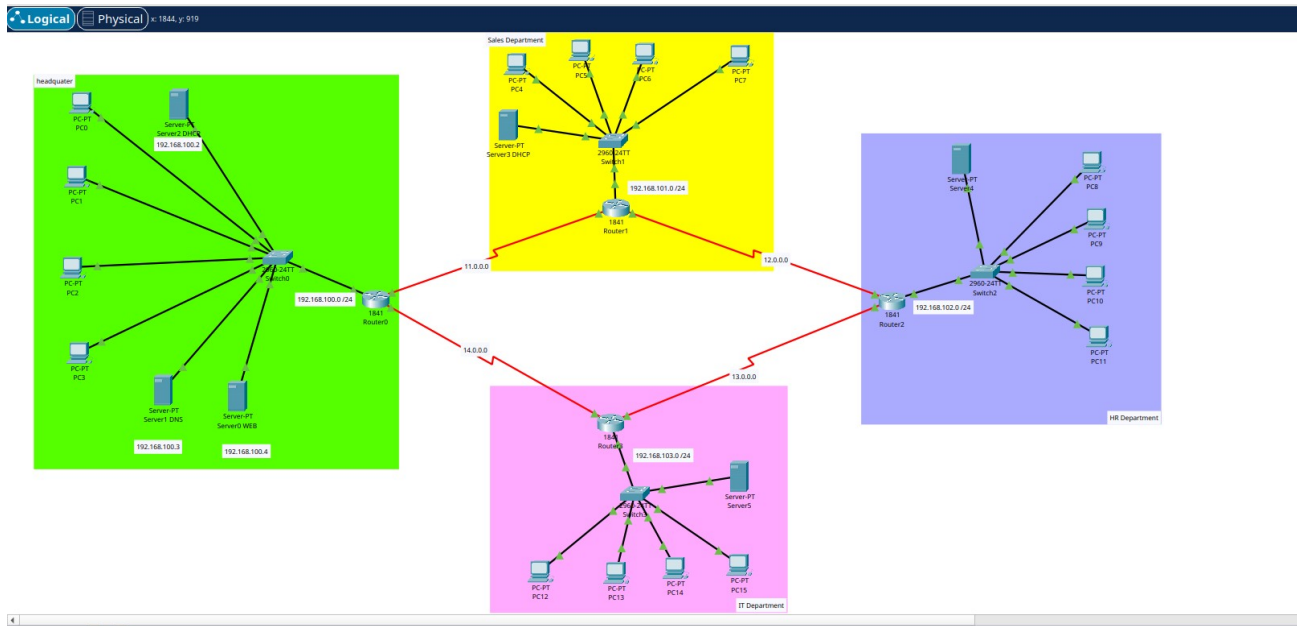
Departments: Headquarters | Sales | HR | IT

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1. Project Overview

This report describes a multi-branch office network designed and simulated in Cisco Packet Tracer. The topology connects four sites — Headquarters, Sales Department, HR Department, and IT Department — through a central hub-and-spoke WAN architecture using Cisco 1841 routers and 2960-24TT switches. Each department has its own LAN subnet, dedicated DHCP server, and end-user PCs. All branch-to-branch and branch-to-HQ communication is routed through Router0 at the headquarters.



2. Network Topology

2.1 Sites and Departments

Site	Color Zone	Router	Switch	LAN Network	End Devices	Servers
Headquarters	Green	Router0 (1841)	Switch0 (2960)	192.168.100.0/24	PC0, PC1, PC2, PC3	Server0 WEB, Server1 DNS, Server2 DHCP
Sales Dept.	Yellow	Router1 (1841)	Switch1 (2960)	192.168.101.0/24	PC4, PC5, PC6, PC7	Server3 DHCP
HR Dept.	Blue/Purple	Router2 (1841)	Switch2 (2960)	192.168.102.0/24	PC8, PC9, PC10, PC11	Server4
IT Dept.	Pink	Router3 (1841)	Switch3 (2960)	192.168.103.0/24	PC12, PC13, PC14, PC15	Server5

2.2 WAN Links (Hub-and-Spoke)

Router0 (HQ) serves as the central hub. Three WAN serial links connect it to each branch router. All WAN links use Class A addresses with a /8 subnet mask (255.0.0.0). Each link uses .2 and .3 as the interface IPs.

WAN Link	Network	Router0 IP	Branch Router IP	Subnet Mask
HQ <-> Sales (Router1)	11.0.0.0	11.0.0.2	11.0.0.3	255.0.0.0 (/8)
HQ <-> HR (Router2)	12.0.0.0	12.0.0.2	12.0.0.3	255.0.0.0 (/8)
HQ <-> IT (Router3)	13.0.0.0	13.0.0.2	13.0.0.3	255.0.0.0 (/8)
HQ <-> (4th link)	14.0.0.0	14.0.0.2	14.0.0.3	255.0.0.0 (/8)

Note: WAN links are represented by red serial cables in the Packet Tracer logical view. No subnetting is applied; the full Class A default mask is used on each point-to-point link.

3. IP Addressing Plan

3.1 Headquarters LAN (192.168.100.0/24)

Device	IP Address	Role
Router0 (LAN interface)	192.168.100.1	Default gateway for HQ LAN
Server2 DHCP	192.168.100.2	Provides DHCP to HQ hosts
Server1 DNS	192.168.100.3	DNS resolution for all sites
Server0 WEB	192.168.100.4	Web server (HTTP)
PC0 – PC3	192.168.100.x (DHCP)	End-user hosts
Switch0	192.168.100.x (mgmt)	Layer 2 access switch

3.2 Sales Department LAN (192.168.101.0/24)

Device	IP Address	Role
Router1 (LAN interface)	192.168.101.1	Default gateway for Sales LAN
Server3 DHCP	192.168.101.x	Provides DHCP to Sales hosts
PC4 – PC7	192.168.101.x (DHCP)	End-user hosts

3.3 HR Department LAN (192.168.102.0/24)

Device	IP Address	Role
Router2 (LAN interface)	192.168.102.1	Default gateway for HR LAN
Server4	192.168.102.x	Local server (file/application)
PC8 – PC11	192.168.102.x (DHCP)	End-user hosts

3.4 IT Department LAN (192.168.103.0/24)

Device	IP Address	Role
Router3 (LAN interface)	192.168.103.1	Default gateway for IT LAN
Server5	192.168.103.x	Local server
PC12 – PC15	192.168.103.x (DHCP)	End-user hosts

4. Devices Inventory

Device Name	Type	Model	Location	Function
Router0	Router	Cisco 1841	Headquarters	HQ gateway, WAN hub, routing
Router1	Router	Cisco 1841	Sales Dept.	Branch gateway, inter-site routing
Router2	Router	Cisco 1841	HR Dept.	Branch gateway, inter-site routing
Router3	Router	Cisco 1841	IT Dept.	Branch gateway, inter-site routing
Switch0	Switch	Cisco 2960-24TT	Headquarters	LAN access switching
Switch1	Switch	Cisco 2960-24TT	Sales Dept.	LAN access switching
Switch2	Switch	Cisco 2960-24TT	HR Dept.	LAN access switching
Switch3	Switch	Cisco 2960-24TT	IT Dept.	LAN access switching
Server0	Server	Generic PT	Headquarters	WEB (HTTP) server
Server1	Server	Generic PT	Headquarters	DNS server
Server2	Server	Generic PT	Headquarters	DHCP server for HQ
Server3	Server	Generic PT	Sales Dept.	DHCP server for Sales
Server4	Server	Generic PT	HR Dept.	Local server
Server5	Server	Generic PT	IT Dept.	Local server
PC0–PC3	PC	PC-PT	Headquarters	End-user hosts (4 units)
PC4–PC7	PC	PC-PT	Sales Dept.	End-user hosts (4 units)
PC8–PC11	PC	PC-PT	HR Dept.	End-user hosts (4 units)
PC12–PC15	PC	PC-PT	IT Dept.	End-user hosts (4 units)

5. Network Services

5.1 DHCP

Dynamic Host Configuration Protocol (DHCP) is deployed to automatically assign IP addresses to end-user PCs.

- Server2 at HQ provides DHCP leases to PC0–PC3 on the 192.168.100.0/24 network
- Server3 at the Sales Department provides DHCP leases to PC4–PC7 on the 192.168.101.0/24 network
- HR and IT departments may use static assignment or their local servers for DHCP
- Each DHCP pool is configured with the appropriate subnet, default gateway (router LAN IP), and DNS server (Server1 IP at HQ)

5.2 DNS

Server1 at Headquarters is configured as the DNS server for the entire network. Its IP address (192.168.100.3) is distributed to all DHCP clients as the preferred DNS server, allowing hostname resolution across all departments.

5.3 WEB Server

Server0 at Headquarters hosts an HTTP web service (192.168.100.4). This server can be accessed by any host in the network (HQ, Sales, HR, IT) through proper inter-site routing, and is used primarily to test end-to-end WAN connectivity and routing configuration.

5.4 Routing

Static routes or a dynamic routing protocol (such as RIP v2 or OSPF) are configured on all four routers to enable communication across all four sites. Router0 acts as the central hub:

- Each branch router (Router1, Router2, Router3) has a default/static route pointing to Router0 for inter-site and internet-bound traffic
- Router0 has static routes to each branch LAN (192.168.101.0/24, 192.168.102.0/24, 192.168.103.0/24) via the respective WAN interface IPs
- WAN interfaces use Class A networks (11.0.0.0, 12.0.0.0, 13.0.0.0, 14.0.0.0) with default mask 255.0.0.0

6. Connectivity Test Results

The following tests were conducted using the Packet Tracer ping utility to verify full end-to-end reachability across all sites:

Source	Destination	Path	Expected Result
PC0 (HQ)	Server0 WEB (HQ)	Local LAN	Success – same subnet
PC0 (HQ)	PC4 (Sales)	HQ LAN → Router0 → 11.0.0.0 → Router1 → Sales LAN	Success – inter-site routing
PC0 (HQ)	PC8 (HR)	HQ LAN → Router0 → 12.0.0.0 → Router2 → HR LAN	Success – inter-site routing

Source	Destination	Path	Expected Result
PC0 (HQ)	PC12 (IT)	HQ LAN → Router0 → 13.0.0.0 → Router3 → IT LAN	Success – inter-site routing
PC4 (Sales)	PC8 (HR)	Sales → Router1 → Router0 → Router2 → HR	Success – via HQ hub
PC8 (HR)	Server0 WEB	HR → Router2 → Router0 → HQ LAN	Success – web access
Any PC	Server1 DNS	Via routing path to 192.168.100.3	Success – DNS resolution
PC12 (IT)	PC15 (IT)	Local LAN (Switch3)	Success – same subnet

7. Summary

The Multi-Branch Office Network project simulates a real-world enterprise network with four departmental sites connected in a hub-and-spoke topology. The key design elements are:

- Four sites: Headquarters, Sales Department, HR Department, and IT Department
- Each site has a dedicated Cisco 1841 router, Cisco 2960-24TT switch, 4 PCs, and at least one server
- LAN networks use Class C private addresses: 192.168.100.0–103.0/24 (one per site)
- WAN links between HQ (Router0) and each branch use Class A networks (11.0.0.0–14.0.0.0) with 255.0.0.0 masks and .2/.3 interface IPs
- DHCP is provided locally at HQ (Server2) and Sales (Server3)
- DNS is centralized at HQ (Server1 at 192.168.100.3)
- A WEB server (Server0 at 192.168.100.4) is accessible from all sites
- All inter-site routing is handled through Router0 at HQ acting as the central hub

The simulation confirms that all hosts can communicate with each other and access shared network services, validating the correctness of the IP addressing, routing, and service configurations.

— End of Report —